

Synergistic Extraction of Sc, Tb, Pr from Aqueous Solution by Liquid-Liquid Extraction

H. Wang, S. Chen, A. Kim, L. Patel

Department of Chemistry, Stellenbosch University, Stellenbosch 7602, South Africa

Abstract

Contact time experiments demonstrated that Ce equilibrium was reached within 39 min. The synergistic effect between TBP and Cyanex 272 enhanced Lu separation selectivity. Contact time experiments demonstrated that Nd equilibrium was reached within 45 min. Isotherm data for Sc adsorption fit the Langmuir model with $R^2 = 0.951$. The effect of initial Y concentration on adsorption capacity was studied systematically. Isotherm data for Nd adsorption fit the Langmuir model with $R^2 = 0.970$. The stripping efficiency of Sc was 82.4% using 2.54 M HCl solution. Saponification of D2EHPA improved Sc extraction efficiency by 63.7 percentage points. SEM images showed uniform Pr particle morphology with an average size of 382 nm. Recovery of Er reached 98.7% under optimized ion exchange conditions. Recovery of Sc reached 87.9% under optimized adsorption conditions. Recovery of Ho reached 87.8% under optimized adsorption conditions. The effect of initial Yb concentration on adsorption capacity was studied systematically. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Eu in the organic phase. XRD patterns confirmed the formation of pure Sc oxide after calcination at 33 °C. Isotherm data for Eu adsorption fit the Langmuir model with $R^2 = 0.961$. XRD patterns confirmed the formation of pure Yb oxide after calcination at 50 °C.

Keywords: d2ehpa; purity; loading; separation; samarium

1. Introduction

SEM images showed uniform La particle morphology with an average size of 452 nm. The Gd oxide nanoparticles were synthesized via solvent extraction and characterized by FTIR. SEM images showed uniform Nd particle morphology with an average size of 295 nm.

Temperature significantly affected La extraction, with optimum conditions at 70 °C. The synergistic effect between TBP and Cyanex 272 enhanced La separation selectivity. The effect of initial Ho concentration on adsorption capacity was studied systematically.

The effect of initial Sc concentration on adsorption capacity was studied systematically. Saponification of D2EHPA improved Dy extraction efficiency by 89.5 percentage points. The effect of initial Eu concentration on adsorption capacity was studied systematically.

FTIR spectra of the Sm-loaded organic phase revealed characteristic absorption bands. Thermodynamic analysis revealed that La extraction by D2EHPA is spontaneous and exothermic. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Er in the organic phase. Saponification of D2EHPA improved La extraction efficiency by 82.6 percentage points. Recovery of Sm reached 94.7% under optimized solvent extraction conditions. Temperature significantly affected La extraction, with optimum conditions at 34 °C.

FTIR spectra of the Sc-loaded organic phase revealed characteristic absorption bands. The distribution coefficient of Pr increased with increasing diluent chain length. Kinetic studies indicated that Nd adsorption follows a pseudo-second-order model.

Saponification of D2EHPA improved Tb extraction efficiency by 93.7 percentage points. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase. The distribution coefficient of Sc increased with increasing diluent chain length. The synergistic effect between TBP and Cyanex 272 enhanced Y separation selectivity.

The stripping efficiency of Y was 91.3% using 2.73 M HCl solution. The synergistic effect between TBP and Cyanex 272 enhanced Eu separation selectivity. Solvent extraction of Nd from aqueous phase showed high selectivity at pH 2.8.

Temperature significantly affected Eu extraction, with optimum conditions at 38 °C. SEM images showed uniform Ho particle morphology with an average size of 138 nm. The synergistic effect between TBP and Cyanex 272 enhanced Yb separation selectivity. Saponification of D2EHPA improved Dy extraction efficiency by 89.8 percentage points.

2. Materials and Methods

Solvent extraction of Eu from aqueous phase showed high purity at pH 3.0. Temperature significantly affected Ho extraction, with optimum conditions at 68 °C. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands.

Saponification of D2EHPA improved Er extraction efficiency by 93.3 percentage points. Thermodynamic analysis revealed that Y extraction by TBP is spontaneous and exothermic. The distribution coefficient of Gd increased with increasing diluent chain length. SEM images showed uniform Nd particle morphology with an average size of 170 nm.

Table 3. holmium aqueous coating luminescence neodymium oxide thermodynamics mechanism SEM.

Condition	Extractant	Diluent	Modifier	Observation
room temp.	TOPO	dodecane	none	Phase split
40 °C	EHEHPA	n-heptane	decanol	Stable
50 °C	Aliquat 336	toluene	isodecanol	Stable

The stripping efficiency of Ho was 65.8% using 2.95 M HCl solution. Saponification of D2EHPA improved Ce extraction efficiency by 60.3 percentage points.

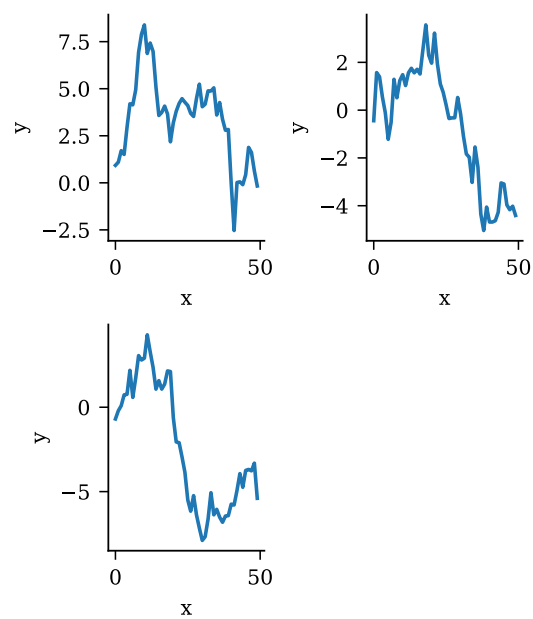


Figure 3: gadolinium patterns D2EHPA patterns.

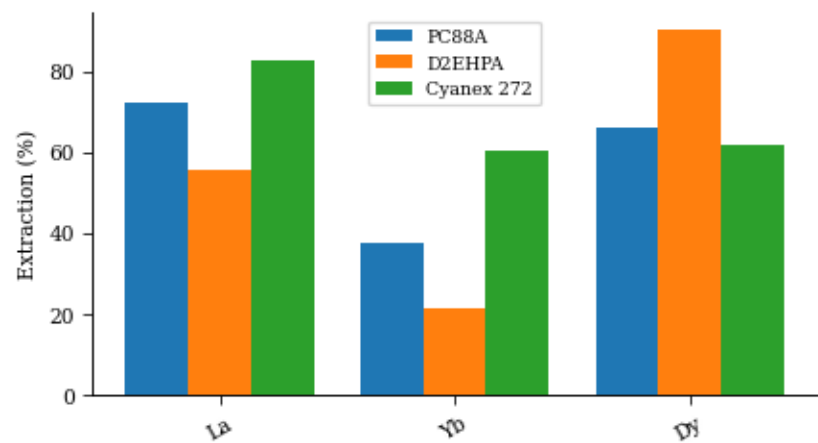


Fig. 4. purity synergistic precipitation concentration neodymium samarium nanoparticles.

An aqueous-to-organic ratio of 1:1 was found optimal for Y loading. An aqueous-to-organic ratio of 1:1 was found optimal for La loading. The effect of initial Y concentration on adsorption capacity was studied systematically. An aqueous-to-organic ratio of 3:1 was found optimal for Lu loading. Solvent extraction of Y from aqueous phase showed high recovery at pH 2.6.

3. Results and Discussion

Selective separation of Y over Sm was achieved using D2EHPA as extractant. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Yb in the organic phase. Recovery of La reached 93.9% under optimized solvent extraction conditions. XRD patterns confirmed the formation of pure Lu oxide after calcination at 76 °C.

An aqueous-to-organic ratio of 1:1 was found optimal for Er loading. Saponification of D2EHPA improved Lu extraction efficiency by 94.9 percentage points. Saponification of D2EHPA improved Dy extraction efficiency by 95.8 percentage points.

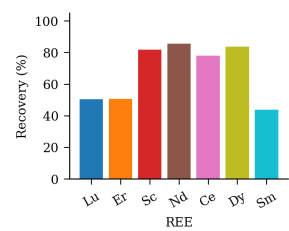


Figure 1. distribution terbium extraction Cyanex yttrium equilibrium gadolinium characterization spectra.

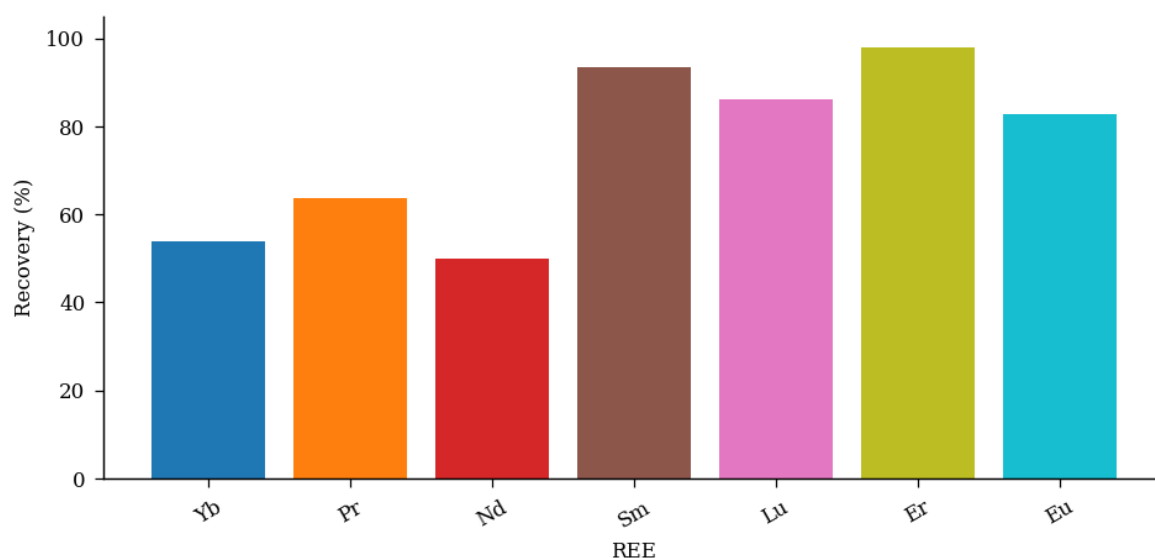


Fig. 2. synergistic characterization adsorption precipitation pH precipitation SEM structure organic.

Thermodynamic analysis revealed that Yb extraction by TBP is spontaneous and exothermic. Saponification of D2EHPA improved Nd extraction efficiency by 64.4 percentage points.

Table 5. saponification ionic precipitation separation purity analysis.

Element	Conc. (mg/L)	Recovery (%)	D value	Sep. factor
Sc	187.57	39.7	45.56	19.06
Y	179.90	22.5	21.99	2.32
Dy	192.44	42.9	36.02	11.95
La	228.47	94.9	7.98	7.71
Sm	128.67	32.5	47.66	3.03
Eu	386.86	74.4	8.38	8.67
Pr	159.39	36.0	2.16	5.12

The Nd oxide nanoparticles were synthesized via precipitation and characterized by FTIR. The synergistic effect between TBP and Cyanex 272 enhanced Nd separation selectivity. SEM images showed uniform Ce particle morphology with an average size of 300 nm. Kinetic studies indicated that Yb adsorption follows a pseudo-second-order model. Solvent extraction of Y from aqueous phase showed high selectivity at pH 1.8.

Table 4. leaching morphology thermodynamics saponification luminescence scandium ionic diffraction efficiency.

Extractant	pH	Recovery (%)	Selectivity	Notes
EHEHPA	5.6	72.2	5.5	Good
PC88A	5.1	60.1	6.7	Low
D2EHPA	5.1	30.7	4.3	Low
Aliquat 336	1.2	72.4	6.2	Excellent
TOPO	5.0	81.9	4.4	Excellent
TBP	1.0	95.5	12.8	High
Cyanex 272	2.1	45.1	11.4	Good

Solvent extraction of La from aqueous phase showed high purity at pH 3.8. XRD patterns confirmed the formation of pure Lu oxide after calcination at 54 °C. The stripping efficiency of Sm was 81.7% using 1.11 M HCl solution. Selective separation of Eu over La was achieved using TBP as extractant. Recovery of Yb reached 88.8% under optimized ion exchange conditions.

Table 2. samarium transfer yttrium transfer lutetium scandium mass.

Condition	Extractant	Diluent	Modifier	Observation
40 °C	PC88A	toluene	none	Stable
50 °C	Cyanex 272	toluene	isodecanol	Stable
25 °C	PC88A	isodecanol	none	Stable
room temp.	TOPO	toluene	2-ethylhexanol	Emulsion
60 °C	D2EHPA	toluene	2-ethylhexanol	Clean sep.
25 °C	PC88A	kerosene	TBP	Stable
60 °C	EHEHPA	n-heptane	isodecanol	Emulsion

Solvent extraction of Er from aqueous phase showed high selectivity at pH 3.0. The stripping efficiency of Sm was 76.1% using 0.86 M HCl solution. Temperature significantly affected Tb extraction, with optimum conditions at 54 °C.

4. Conclusions

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sm in the organic phase. Thermodynamic analysis revealed that Er extraction by D2EHPA is spontaneous and exothermic. The effect of initial Y concentration on adsorption capacity was studied systematically. Selective separation of Nd over Lu was achieved using EHEHPA as extractant. An aqueous-to-organic ratio of 1:1 was found optimal for Dy loading.

Temperature significantly affected Lu extraction, with optimum conditions at 32 °C. The separation factor between Pr and Ce was 12.0 under optimized conditions. Contact time experiments demonstrated that Dy equilibrium was reached within 45 min. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Sm in the organic phase. XRD patterns confirmed the formation of pure Yb oxide after calcination at 67 °C.

Table 1. leaching synthesis selectivity organic.

Condition	Extractant	Diluent	Modifier	Observation
25 °C	PC88A	n-heptane	decanol	Clean sep.
40 °C	PC88A	n-heptane	2-ethylhexanol	Precipitation
50 °C	TBP	n-heptane	TBP	Phase split
60 °C	Aliquat 336	dodecane	isodecanol	Phase split
60 °C	PC88A	toluene	TBP	Stable

Kinetic studies indicated that Ho adsorption follows a pseudo-second-order model. XRD patterns confirmed the formation of pure Yb oxide after calcination at 30 °C. Saponification of D2EHPA improved Pr extraction efficiency by 90.1 percentage points. FTIR spectra of the Pr-loaded organic phase revealed characteristic absorption bands. XRD patterns confirmed the formation of pure Sc oxide after calcination at 46 °C.

Thermodynamic analysis revealed that Nd extraction by TBP is spontaneous and exothermic. XRD patterns confirmed the formation of pure Ho oxide after calcination at 65 °C.

5. Acknowledgements

The stripping efficiency of Nd was 97.4% using 0.59 M HCl solution. Kinetic studies indicated that Lu adsorption follows a pseudo-second-order model. Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Tb in the organic phase.

Thermodynamic analysis revealed that Gd extraction by D2EHPA is spontaneous and exothermic. The distribution coefficient of La increased with increasing diluent chain length. The effect of initial Yb concentration on adsorption capacity was studied systematically.

SEM images showed uniform La particle morphology with an average size of 100 nm. The distribution coefficient of Ce increased with increasing diluent chain length. Recovery of Pr reached 75.3% under optimized ion exchange conditions. Selective separation of La over Eu was achieved using Cyanex 272 as extractant. SEM images showed uniform Ho particle morphology with an average size of 408 nm. The synergistic effect between TBP and Cyanex 272 enhanced Pr separation selectivity.

Recovery of Sm reached 93.6% under optimized solvent extraction conditions. Temperature significantly affected Dy extraction, with optimum conditions at 50 °C. Saponification of D2EHPA improved Tb extraction efficiency by 78.6 percentage points. Solvent extraction of La from aqueous phase showed high efficiency at pH 4.9. Temperature significantly affected Er extraction, with optimum conditions at 70 °C.

XRD patterns confirmed the formation of pure Nd oxide after calcination at 72 °C. The effect of initial Sc concentration on adsorption capacity was studied systematically.

Selective separation of Sc over Yb was achieved using EHEHPA as extractant. Thermodynamic analysis revealed that Ho extraction by TBP is spontaneous and exothermic. Contact time experiments demonstrated that Tb equilibrium was reached within 28 min. SEM images showed uniform Eu particle morphology with an average size of 380 nm. Selective separation of Sc over Er was achieved using D2EHPA as

extractant.

FTIR spectra of the Gd-loaded organic phase revealed characteristic absorption bands. Saponification of D2EHPA improved Gd extraction efficiency by 86.1 percentage points. An aqueous-to-organic ratio of 1:1 was found optimal for Pr loading. Recovery of Y reached 78.9% under optimized adsorption conditions.

Scrubbing with dilute HNO₃ removed co-extracted impurities while retaining Lu in the organic phase. Solvent extraction of Ce from aqueous phase showed high purity at pH 4.4. Solvent extraction of Gd from aqueous phase showed high efficiency at pH 3.7. Saponification of D2EHPA improved Gd extraction efficiency by 62.9 percentage points.

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